MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, PERIPHERAL TYPE, MISCELLANEOUS

NAVAL SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements applicable to the design and construction of peripheral pumps for miscellaneous services onboard Naval ships.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein,

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187 - Bearings, Roller, Tapered.
QQ-C-390 - Copper Alloy Castings (Including Cast Bar),
QQ-N-281 - Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
GGG-P-781 - Puller, Mechanical Puller Attachment, Mechanical, and Puller Set, Mechanical.

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact) ; Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment Basic Requirements, (Naval Shipboard Use).
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C., and Associated Switching Devices, Naval Shipboard.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-S-15083 - Steel Caskings.
MIL-T-16420 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Seamless and Welded.
MIL-P-16709 - Packaging of Pumps Including Prime Movers and Associated Repair Parts.
MIL-M-17059 - Motiors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use).
MIL-M-17060 - Motors, 60-Hertz, Alternating Current, Integral-Horsepower Shipboard Use.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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MIL-M-17413 - Motors, Direct Current, Integral H.P. , Naval Shipboard.
MIL-M-17556 - Motor, Direct-Current, Fractional Hp. (Shipboard Use).
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).

STANDARDS

MILITARY

MXL-STD-129 - Marking For Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of shipboard Equipment (Type I - Environmental and Type II - Internally Excited) .
MIL-STD-278 - Fabrication, Welding and InspectIon; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Submarine Service.
MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships.

DRAWING

NAVSEA

810-L385850 - Piping Instruments, Pressure for All Service.

PUBLICATIONS

NAVSEA 0900-LP-001-7000 - Piping Systems Brazed, Fabrication and Inspection.
NRL Report 7396 - Shipboard Shock and Navy Devices For Its Simulation.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications, The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)
Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office Washington D.C. 20402) .

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materia1s. Materials shall conform to table I. Proposed alternative materials shall give equal or better service than the materials specified. Proposed alternative matarials shall be subject to review by the design review activity. Components of the pump for which specific materials are not specified shall be materials compatible for the service intended, Materials which can be sensitized and are subjected to heat treatment in the sensitization range during fabrication shall be capable of passing the intergranular corrosion tests of the base material specification. Materials shall not be sensitized during hard facing,stress relieving, or repair welding.

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TABLE I. Materials for pump.

Application	Material	Specification
Casings	Gun metal	QQ-C-390, alloy 903
	Valve bronze	QQ-C-390, alloy 922
Casing bolts	Ni-Cu alloy rolled	QQ-N-281
Impellers	Ni-Cu alloy cast	QQ-N-288
	Gun metal	QQ-C-390, alloy 903
	Valve bronze	QQ-C-390, alloy 922
Shafts	Ni-Cu-Al alloy	QQ-N-286
	Ni-Cu alloy rolled	QQ-N-281
Bearing brackets and caps	Gun metal	QQ-C-390, alloy 903
	Valve bronze	QQ-C-390, alloy 922
Glands	Gun metal	QQ-C-390, alloy 903
	Valve bronze	QQ-C-390, alloy 922
Mounting bracket for close-coupled pumps	Gun metal	QQ-C-390, alloy 903
	Valve bronze	QQ-C-390, alloy 922
Tanks	Ni-Cu alloy	QQ-N-281

3.2 General design. The design and construction of all pumps shall comply with the space and wieght requirements when specified (see 6.2,1). The pump unit shall be designed to perform under the list, pitch, roll and trim conditions as specified (see 6.2,1), Pumps shall be capable of operating continuously at any point over the entire flow range and design conditions as specified (see 6.2.1).

3.2.1 Reliability. The assurance of maximum reliability shall be a paramount controlling principle in the design, fabrication, assembly, and testing of this equipment, Users of this equipment and contractors are urged to advise the Naval Ship Engineering Center (NAVSEC) of any findings whereby improvement of equipment reliability can be achieved, Equipment specified herein is intended to be operated, maintained, and repaired onboard Navy ships. Attention is directed to the high equipment density of shipboard machinery spaces, to the fact that maintenance and repairs must be made underway in heavy seas, that equipment must operate unattended, and that maintenance personnel may not be seasoned mechanics. These factors shall be considered in the pump design, Aspects of "human engineering" shall also be fully explored and considered to minimize the possibility of failure through improved operation and maintenance procedures and to preclude occurrence of personnel safety hazards. Parts shall be physically interchangeable or reversible when such parts are also interchangeable functionally and from considerations of strength.

3.2.2 Performance characteristics. The operating characteristics of the pumps and drivers shall be as specified (see 6.2.1).

3.2.2.1 Horizontal flexible-coupled pump and separable driving Unit, complete with all appurtenances, shall be mounted on a common bedplate. Bedplates shall not be required for close-coupled type pumps.

3.2.2.2 Bedplates shall be of cast steel, grade CW, of MIL-S-15083 or of structural steel fabricated by welding.

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3.2.2.3 Bedplates shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship without disturbing the alinement of the assembled units and such that the normal distortion weaving, or vibration of the supporting structures onboard ships cannot cause misalinement between the pumps and driving units.

3.2.2.4 Bearing and seating surfaces of bedplates shall be finish machined.

3.2.2.5 Component parts of an assembled unit supported directly by a bedplate shall be doweled thereto to facilitate reassembly and maintenance of alinement.

3.2.3 Bearings. Radial and thrust bearings shall be of the rolling contact type in l ccordance with PF-B-171, FF-B-185, or FF-B-1E7, as applicable.

3.2.3.1 Installation.

3.2. 3.1.1 Bearings of flexible-coupled pumps shall be installed in housings separate from and independent of the stuffing boxes and with provision for protection from gland leakage. Means shall be provided to prevent escape of lubricant around the shaft. Baffles, wipers, and related parts shall be readily renewable.

3.2,3.1.2 Bearing brackets shall be either cast integral with the casing or securely bolted to the casing on a machined shoulder, and held in alinement with the casing by at least two heavy dowels. Bolts shall not be used alone for securing brackets.

3.2. 3.1.3 Bearing housings shall be cast integral with the bracket or secured thereto in such manner as to insure alinement.

3.2.3.2 Thrust bearings.

3.2.3.2.1 Thrust bearings shall be used in each flexible-coupled pump for counteratting any unbalanced hydraulic or mechanical thrust in either direction. Rolling, pitching, or list of a ship shall be included in determinantion of total thrust loads.

3.2. 3.2.2 Thrust bearings may be of the combined radial and thrust rolling contact type.

3.2.4 Lubrication. Lubrication of ball bearings shall be in accordance with FF-B-171, FF-B-185, or FF-B-187.

3.2.5 Piping and valves. Valves, flanges, and fittings for pipe connections shall conform to

3.2.6 Priming shall be accomplished by the means of a suction trap which shall contain l strainer basket and chock valve.

3.2.7 Motors.

3.2.7.1 Motors for driving pumps shall be of the types and characteristics specified (see 6.2.1), and shall conform to MIL-M-17413 or MIL-M-17556 for direct current (d.c.) motors or MIL-M-17059 or MIL-M-17060 for alternating current (a.c,) motors.

3.2.7.2 The power rating of motors with ratings less than two brake horsepower (bhp) at rated conditions shall be the next Navy standard rating larger than the rated bhp multiplied by a safety factor of 1.5 - (Rated bhp).
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3.2.7.3 The power rating of motors with bhp ratings of two or more bhp shall be not less than 110 percent of the bhp of the driven pump at maximum rated condition.

3.2.8 Controllers. Motor controllers, including pressure switches, shall have the charactori8kstics specified (see 6.2.1) and shall conform to MIL-C-2212.

3.2.9 Pressure tanks.

3.2.9.1 Tank for packaged water systems shall be of the size specified (see 6.2.1) and shall be equippd with an automatic air volume control and pressure gage. Tanks shall comply with the test requirements of 4.3.2.1.

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3.2.9.2 Packaged systems equipped with pressure tanks and pumps shall be furnished with check valves and all piping and fittings required between the pump and tank. Piping, valves and fittings shall be in accordance with 3.2.5.

3.2.10 Gages. Pressure gages, when provided with the units shall be mounted in accordance with Drawing 810-1385850.

3.2.11 Shock and vibration.

3.2.11.1 Shock. Pumping units with all appurtenances and controls, shall be capable of passing the class HI (high-impact) shock test specified in MIL-S-901 (see 4.3.3).

3.2.11.1.1 Shockmounts shall not be used unless command or agency approval *of* the mount and its proposed application is given.

3.2.11.1.2 Bolts shall be installed in holes *no* greater than the following sizes:

Nominal bolt diameter (Inches)	Maximum diameter of hole (Inches)
3/4 and smaller	Nominal bolt diameter plus 1/32 inch.
Larger than 3/4	Nominal bolt diameter plus 1/16 inch.

3.2.11.2 Vibration. The units shall comply with the requirements of MIL-STD-L67-1, type I environmental vibration up to and including 33 Hz. *The* units shall be tested under nominal load conditions to duplicate shipboard installation. Submission of a detailed teat "Set up" shall be submitted prior to any testing. Test reports shall be in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.2.12 Casings.

3.2.12.1 The casings of flexible-coupled pumps shall permit ready removal of the rotors and replacement of bearings and glands.

3.2.12,2 The casings of close-coupled pumps shall permit ready replacement of wearing parts.

3.2.12.3 Clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools.

3.2.12.4 Pumps shall be preferably of the self-venting type with suction and discharge at the top of the pump casing. If design is not self-venting, casing shall be provided with vent connection at high point.

3.2.13 Suction and discharge connections.

3.2.13.1 Suction and discharge connections shall be flanged,

3.2.13.2 Suction and discharge connections shall, in general, be located at the top of casings and positioned so that the pump can be disassembled without breaking piping connections.

3.2.13.3 Suction and discharge connections shall conform to the requirements of MIL-F-20042, MIL-STD-438, and MIL-STD-777, as applicable.

3.2.14 Impellers and shafts.

3.2.14.1 Surfaces of impellers shall be smooth finished with impeller vanes accurately milled on both sides to uniform thickness and forming uniformly dimensioned buckets between the vanes. The vanes on the two sides of the impeller shall be staggered so that each vane will be approximately centrally located with reference to two vanes on opposite side of the **impeller.**

3.2.14.2 Impellers shall have pressure balancing openings between the inner diameter of sealing ring and the hub of the impeller,

3.2.14.3 Impellers shall be keyed on the shaft.

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3.2.14.4 Shaft threading shall be counter to the direction of rotation except where both right hand and left hand units of the same design are involved; in which case, in order to reduce the number of shafts required for repair parts, the threading shall be right hand for all units and the threaded parts shall be locked in place.

3.2.14.5 Shaft flingers shall be provided adjacent to stuffing boxes for all pumps.

3.2.14.6 Shafts of d.c. motor driven pumps shall be provided with means to permit the ready and direct use of portable tachometers.

3.2.15 Stuffing boxes and packing.

3.2.15.1 Pumps shall be provided with mechanical shaft seals. Pumps for salt water service shall also be fitted with cyclone separators (see 6.2.1) . Cyclone separators shall be constructed of nickel-copper alloy. The gland of each seal shall incorporate two or more rings of packing for use in the event of a mechanical seal failure, Seal internals for sea water service pumps shall be constructed of nickel-copper alloy. -Seal internals for fresh water service pumps shall be constructed of 304 or 316 corrosion-resistant steel or Ni-Cu alloy. The mechanical shaft seal shall have a tungsten carbide against carbon sealing surface. Fittings for abrasive separators and pump casings shall be of the straight thread type with O-ring seals and shall be constructed of nickel-copper alloy. Tubing shall be copper-nickel (70-30) In accordance with MIL-T-16420.

3.2.15.2 Mechanical shaft seals shall be positioned on the shaft by means of stub or step sleeves. Mechanical shaft seals shall not be positioned by use of set screws alone.

3.2.15.3 Positive liquid pressure shall be supplied to the seal faces under all conditions of oprration and adequate circulation of the liquid at the seal faces shall be provided to minimize the deposit of foreign matter on the seal parts.

3.2.15.4 Stuffing box glands shall be set up by nuts threaded on studs secured in the casings.

3.2.15.5 Space shall be provided between bearings and stuffing boxes to permit easy inspection of mechanical seals and bearings.

3.2.16 Couplings.

3.2.16.1 Flexible-couplings shall be of all-metal construction.

3.2.16.2 Flexible-coupling hubs shall be keyed to the shafts. For shafts 3/4 inch and smaller, the coupling hubs shall be secured to the shafts by lock nuts or set screws. For shafts larger than 3/4 inch, the coupling hub shall be secured to the shafts by lock nuts. For shafts 1 inch in diameter and larger, the hubs shall be fitted on a taper with keys parallel to the taper.

3.2.16.3 Flexibly-coupled units shall have coupling guards. Guards shall permit ready access to the coupling for lubrication and inspection.

3.2.17 Fittings,

3.2.17.1 The bottom of all pump casings shall be fitted with drain connections to permit complete draining of the pumps without disassembly of the pumps.

3.2.17.2 The bottom of drip pockets shall be pipe tapped for drain connections.

3.2.18 Painting.

3.2.18.1 All external unmachined surfaces that are not inherently corrosion-resistant a s defined in MIL-E-917 shallall be processeed (treaed, plated, or painted) in accordance with MIL-E-917 to provide corrosion-resistnace. Paint shall not be applied to any internal machined surfaces. When desired, (for convenience or appearance) contiguous non-ferrous surfaces may also be painted.

3.2.19 Threaded parts such as bolts, studs, and nuts shall conform to Handbook H28.

3.2.20 Welding shall be in accordance with MIL-STD-278.

3,2.20.1 Where practical, the use of silver-brazed joints shall be avoided. Where used silver-brazed joints shall conform to the fabrication and inspection requirements of NAVSEA 0900-LP-001-7000.

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3.2.21 Pumps shall have constantly rising head capacity curves.

3.3 Identification plates.

3.3.1 Identification plates shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, and shall be of material other than plastic or aluminum.

3.3.2 Identification plates shall be secured to equipment with corrosion-resistant metallic screws.

3.3.3 Pump identification plates shall contain data as follows:

(a) Manufacturer's name
(b) Manufacturer's model or type and size,
(c) Service application.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
  (1) Capacity in gallons per minute (gal/rnin).
  (2) Total head in pounds per square inch (lb/in^2).
  (3) Speed of shaft in revolutions per minute (r/rein).
  (4) Brake horsepower (bhp).
  (5) Test pressure.
  (6) Special data vital to the unit:
    (a.) Suction pressure.
    (b.) Submergence.
(f) Contract number (and item number for multiple unit orders).
(g) National stock number.
(h) Section for Defense Contract Administration Services (DCAS) stamp.

3.3.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.4 Repair parts and tools.

3.4.1 Onboard. Onboard repair parts of the types and quantities shown in table II shall be furnished as required (see 6.2.1).

TABLE II. Onboard repair parts.

Name of part	Quantity per set
Bearings Coupling bushings, pins, and other wearing parts Felt wiper rings or oil seals Mechanical seals	100 percent complete replacement for all installed units per ship
Rotors, completely assembled, including shaft, impellers, casing diaphragms and bushings, keys, nuts, flingers, and complete coupling.	100 percent replacement for one pump.

3.4.1.1 The pump rotor assembly for flexible-coupled units shall be completely assembled ready for installation in the unit for which intended and shall include any nonrotating parts which can be installed only by removing some rotating part or parts of the pump rotor.

3.4.1.2 Pump rotor assemblies for motor driven close-coupled pumps shall be furnished disassembled. Shaft shall be furnished with motor rotor or armature.

3.4.1.2.1 The quantities of individual repair parts shall not be reduced because of duplication of parts furnished to make the rotor assembly.

3.4.1.3 Repair parts for the drivers and associated equipments shall be furnished in accordance with the applicable equipment specification,

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3.4.2 Tools.

3.4.2.1 A complete set of special tools required for the maintenance and repair of the pump units shall be included in each set of onboard repair parts. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.4.2.2 The special tools shall include the following, as applicable:

(a) Complete set of special wrenches of forged steel with hardened jaws.
(b) One impeller pulling tool.
(c) One sat of ball bearing pulling tools.
(d) Any other special tools necessary for the proper maintenance of the units in service.

3.4.2.3 Tools which are available in the National Supply Catalog, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-782 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed and Included in the list of material.

3.4.3 Stock. The selection, stock numbering, and quantities of repair parts for stock shall be detjermined and processed in accordance with data ordering document included in the contract or order (see 6.2.2).

3.4.3.1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania.

3.5 Technical data. The contractor shall prepare drawings, manuals, and provisioning list in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.5.1 through 3.5.3.4.

3.5.1 Drawings. In addition to the drawing content required by the data ordering documents (6.2.2), the technical features in 3.5.1.1 through 3.5.1.6 shall be included.

3.5.1.1 Outline drawings shall show complete pump units including pump, driver, bedplate, and attached accessories. There shall be separate outline drawings for each contract or order and for each different pump unit on a contract or order.

3.5.1.2 Pump drawings shall include sectional assembly, complete list of material, and details of all parts. The sectional assembly and list of material with references to detail drawings should be on a single sheet; and details shall be on one or more additional sheets. Several details should be shown on one drawing; details of single pump parts on size A sheets, resulting in a multiplicity of drawing sheets, should be avoided.

3.5.1.3 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the ‘List of Material” of the sectional assembly drawing,

3.5.1.4 Drawings for drivers and associated equipment shall be in accordance with the specificationa for those equipment.

3.5.1.5 Sectional assembly drawing shall show all assembly clearances.

3.5.1.6 Drawings submitted with bids arenot required to be in drawing format.

3.5.2 Provisioning parts list. In addition to the provisioning parts list content required by the data ordering documents (see 6.2.2), the features specified in 3.5.2.1 through 3.5.2.3 shall be included.

3.5.2.1 Each box containing onboard repair parts shall contain a list entitled “List of onboard repair parts and tools”. The list shall be in a format for use on outline drawings and in the instruction books. It shall be not less than nominal 8-1/2 by 11 inches in size.

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3.5.2.2 The format of the list shall include a heading and columns of data for the items listed, The heading shall include titles and applicable entries as follows:

Reproduced from drawing number_____
Number of ships_____
Application_____
Contract number_____
Manufacturer_____
Quantities are for_____units per ship
The columns shall include:
Piece number_____
Name of part_____
Quantity_____
Drawing number_____
National stock number_____

Additional columns may be used as applicable.

3.5.2.3 The list shall be so treated as to be resistant to oil, water, and fading.

3.5.3 Manuals. Unless otherwise specified (see 6.2.1), technical manuals shall be furnished in accordance with the data ordering documents included in the contract or order (see 6.2.2). In addition, the features specified in 3.5.3.1 through 3.5.3.4 shall be included,

3.5.3.1 A separate manual shall be furnished for each different pump unit except in special cases when manuals covering more than one pump may be required. Preliminary manuals shall be submitted to the design, review activity for review and shall include all proposed sections complete. Unless otherwise specified (see 6.2.1), manuals shall include pump, driver, controls and all appurtenances and all applicable drawings. Performance curves for final manuals may be furnished as insert pages after delivery of hardware.

3.5.3.2 Manuals covering pumps driven by conventional electric motors with conventional controllers need not include operation and maintenance instructions covering the motors and controllers. They shall, however, include sectional assembly drawings and certification data covering the motors and controllers. Complete operation and maintenance instructions covering electrical equipment shall be included when specifically required (see 6.2.1).

3.5.3.3 Each manual shall include not less than the following illustrations covering the pump and driver:

(a) Sectional assembly drawing.
(b) Outline drawing.
(c) Complete list of material corresponding to the sectional assembly drawing.
(d) A minimum of two photographs of the complete unit (pump and driver) taken 180 degrees apart on a horizontal plane. Where several sizes of identical design units are being supplied, photographs of one design unit only need be supplied.

3.5.3.4 The alignment procedure in the technical manual shall describe in complete detail the means by which the required alinement clearance at bearings and wearing rings is to be established. The manual shall prescribe the maximum permissible measured eccentricities which can be tolerated in the aforementioned areas with the pump in the assembled condition. Exception to any part of the aforementionod requirements in any specific area may be had only by providing for the design review activity approval a written technical justification for the said exception. The manual shall specify the maximum allowable clearances for wearing rings and sleeve bearings before the replacement of the part(s). In addition the manuals shall contain:

(a) Lubrication schedule, including lubricant and procedure.
{b) Detailerd shaft packing instruction. The minimum allowable leakage rate shall be specified,
(c) Threaded fastener torque values and sequence required to maintain proper seal and alinement.
(d) In-field dynamic bahancing instructions where provisions for balancing in-field are required to maintain proper noise and vibration characteristics.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the

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contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure contractors and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data and ordering documents included in the contract or order (see 6.2.2).

4.1.2 When specified (see 6.2.1) , the pump shall be free of all mercury contamination, During the manufacturing process, tests, and examination, the product to be offered for acceptance shall not come in direct contact with mercury or any of its components nor with any mercury containing device employing a single boundary of containment. The manufacturer shall certify that the product, when shipped, is free from mercury contamination.

4.2 Quality conformance inspection.

4.2.1 Sampling for performance tests.

4,2.1.1 For a packaged tank and pump unit, a representative number of sets in accordance with table III shall be tested to demonstrate the ability of the unit to function properly insofar as the start-stop of the pump, pressure switch, and air volume control are concerned.

TABLE III. Sampling for design-proving tests.

Number of identical units	Number to be tested
2 to 8	3
9 to 15	4
16 to 25	5
26 to 40	6
41 to 65	8
66 to 110	10

4.3 Tests.

4.3.1 Performance.

4.3.1.1 Driving units shall be tested as required by the applicable equipment specifications.

4.3.1.2 One complete pump of each size on order, including the motor and controller, shall be given a test to determine the overall power consumption of the unit under the specified conditions. The Naval Ship Engineering Center (NAVSEC) may consider waiver of overall power consumption tests on pumps if identical units have been previously tested satisfactorily.

4.3.1.2.1 Performance tests shall ordinarily be conducted at the same time as, and shall include the inclined operation tests as specified for the driving unit. For inclined operation tests the unit shall be run at maximum operating speed, but need not be fully loaded.

4.3.1.3 Performance tests shall demonstrate the ability of the pump to handle its rated capacity of specified liquid at the maximum temperature and minimum suction head or maximum suction-lift or vacuum as applicable.

4.3.1.4 The following test curves shall be submitted:

(a) Capacity versus total head.
(b) Capacity versus pump efficiency.
(c) Capacity versus bhp .
(d) Capacity versus electrical horsepower input.

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4.3.2 Shop test,

4.3.2.1 Pump casings and pressure tank, where funished, shall be tested hydrostatically to a gage pressure one and one-half times the maximum discharge pressure or total head, whichever is greater, but in no case to less than SO pounds per square inch (lb/in^2).

4.3.2.2 Pumps driven by d.c. motors shall be tested at full load at 25 percent above designed maximum operating speed by a continuous nonstop run of at least 30 minutes. This test shall be conducted to check for smoothness of operation of pump and motor unit. Pumps driven by a.c. motors shall be tested for smoothness of operation at full load and full speed.

4.3.3 Shock.

4.3.3.1 Unless otherwise specified in the contract or order (see 6.2.1), one complete pumping unit of each type, and size shall be subject to the HI shock tests specified in MIL-S-901 for type A, shock-proofness, class III equipment, except as otherwise required herein.

4.3.3.2 When the shock tests are on a medium-weight shock machine, three blows shall be applied with the pumps mounted in its normal orientation; three blows shall be applied with the pump inclined 30 degrees about one horizontal axis and three blows shall be applied with the pump inclined 30 degrees about the other horizontal axis for a total of nine blows (three blows in each of three groups in accordance with MIL-S-901). The pump shall be tested running and at standstill on alternate blows. Pumps need not be running at rated speed or pumping at rated capacity during the shock test.

4.3.3.3 Equipment exactly duplicating that previously shock tested as required by 4.3.3.1 and 4.3.3.2 and accepted need not be retested except when evidence of low shock resistance develops in installed units. The command or agency concerned will initiate action for correction of deficiencies and reserves the right to require additional shock tests.

4.3.3.4 Pump units shall be shock tested with drivers unless otherwise approved by NAVSEC. Flexible-coupled pumps shock tested with one driver will not be required to be shock tested again when supplied with a different drive f equal or lesser weight, subject to concurrence of the command or agency concerned. In the vent drivers are not shock tested with the pump unit, the drivers will have been or will be shock tested in accordance with MIL-S-901 either separately or with another driver unit.

4.3.3.5 A pump unit shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion, deformation, or dislocation of any part, such as shaft, mounting feet or bearings.
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to tests at rated speed or more than that permitted by the specifications for the driver. The amplitude of unbalance both before and after the shock tests and the speed at which measured shall be recorded in the shock test report (see 4.6).

4.3.3.6 Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial or Government laboratory equipped to conduct tests. Where shock tests are conducted at a Government laboratory, copies of the applicable drawings shall accompany the units.

4.3.3.7 Pump units which have been subjected to the HI shock test and have failed to conform to the requirements will not be acceptable.

4.3.3.8 Pump units which have been subjected to the HI shock test and have successfully passed this test shall be considered acceptable for such service as the command or agency concerned may authorize. Such equipment shall be reconditioned by the manufacturer as follows:

(a) Minor deformations of mounting flanges shall be corrected.
(b) Minor deformations affecting alinement shall be corrected.
(c) All bearings shall be replaced.
(d) Each part shall be carefully examined by the manufacturer end any part which he considers substandard shall be replaced.

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Minor deformations shall be defined as those which do not cause unqualified rejection of the design under HI shock test but which are in excess of the design dimensional tolerances specified on the applicable pump drawings.

4.3.4 Vibration. Unless otherwise specified (see 6.2.1) , 1 pump, of a quantity of identical pumps being supplied, shall be tested in accordance with the requirements of 3.2.11.2.

4.3.5 Mechanical shaft seal and alinement procedure. Mechanical shaft seal and alinesent.procdure test shall be accomplished on one pump of each size and design tested. The mechanical shaft seal shall be installed and alinement don. precisely in accordance with the procedures to be used in the technical manual. The mechanical seal shall be run in and the pump operated at maximum rated speed with the fluid pumped at maximum operating temperature. The tests shall verify seal installation and back-up packing installation procedures Stuffing box leakage, if any, shall be determined. Dynamic balance shall be checked by use of a vibration indicator. The vibration shall not exceed the limits specified in type I of MIL-STD-167-1 for this equipment.

Acceptance criteria: Machanical seal leakage shall not exceed 5 drops per minute.

4.4. Inspection of pump repair parts. The pump repair parts required by 3.4 shall be inspected to assure interchangeability with the similar parts in the assembled pumps. Ordinarily material analyses and physical test speciments will not be required but procedures shall be employed which will assure that the delivery of parts are made of the same materials and by the same processes as were used for the assembled pumps.

4.5 Test reports. The contractor shall prepare test reports and shock test reports in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 4.5.1.

4.5.1 Test reports. In addition to the teat report content required by the data ordering document (see 6.2.2), the unique features specified in 4.5.1.1 and 4.5.1.2 shall be included.

4.5.1.1 Complete test reports, including test data on both pumps and driving units, shall be assembled and distributed as follows:

(a) Three to NAVSEC (to be forwarded via the cognizant Government representative).
(b) One for primary district Defense Contract Administration Service (WAS) representative.
(c) One for branch DCAS (if any).
(d) Two to each Supervisor of Shipbuilding (one for supervisor's files, the other for the shipbuilder).
(e) One to the Commander of each shipbuilding yard concerned, for Navy yard built ships, when specified (ace 6.2.1).

4.5.1.2 Data shall be taken during tests to prepare pump characteristic curves as specified in 4.3.1.4, In all cases the test data shall be corrected to the specified operating conditions as to voltage, frequency, temperatures, specific gravity, suction head or lift, and vacuum, as set forth in the contract or order (see 6.2.1) , such conditions shall be clearly shown on the data sheets. The data sheets shall state the actual finished diameter of the pump impellers installed on the test. Test data and curves shall be complete over the entire range of capacities from the shut-off to as near free delivery as possible.

4.6 Inspection of reparation for delivery. Sample packages and packs and the preservation and packaging, and marking for shipment and storage.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging, packing and marking. Pumps, pump accessories, repair parts, tools and technical data shall be preserved packaged level A or C, packed level A, B or C, as specified (see 6.2.1) and marked in accordance with MIL-P-16789.

5.2 Special requirements. In addition to the special marking requirements of MIL-P-16789, packing lists shall be furnished in accordance with the requirements of MIL-STD-129,

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6. NOTES

6.1 Intended use. Centrifugal, peripheral type pumps are for miscellaneous services onboard Naval ships.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) List, pitch, roll, trim and other ship attitude conditions.
(c) Performance characteristics (see 3.2.2).
  (1) Rated capacity.
  (2) Rated total head at rated capacity.
  (3) Fluid to be pumped (sea water or fresh water) (see 3.2.15.1).
  (4) Specific gravity of fluid,
  (5) Temperature of fluid.
  (6) Suction head available.
  (7) Whether for surface ship or submarine (see 4.3.2.2).
(d) When pump units shall be horizontal close-coupled (see 3.2.2.1).
(e) Electrical characteristics (for motor drive) as required by applicable motor and controller specifications (see 3.2.7.1 and 3.2.8),
(f) Whether pressure tank size required and size when required (see 3.2.9.1).
(g) Number of pumps per ship on which quantity of onboard repair parts is to be based (see 3.4.1).
(h) When technical manuals are not required (see 3.6.3.1).
(i) When appurtenances and all applicable drawings shall be included (see 3.6.4.1).
(j) Mercury prohibition (see 4.1.2).
(k) When Navy yard built ships require test reports (see 4.5.1.1(e)).
(l) Specific gravity of liquid handled, liquid temperature, and suction head (see 4.5.1.2).
(m) Preservation, packaging, packing, and marking required (see 5.1).
(n) Special marking required (see 5.2).
(o) When pump shall be tested for shock and vibration (see 4.3.3.1 and 4.3.4).
(p) When manuals are not required (see 3.5.3).
(q) When complete maintenance and instruction for electrical equipment is required (see 3.5,3.2).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement regulations (ASPR) , the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664) , and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

	Paragraph	Data requirements	Applicable DID	Options
(a)	3.2.11.2 and 4.5 through 4.5.1.2	Test reports	UDI-T-23729	------
(b)	3.4.3 and 3.5.2	Provisioning list	DI-V-2078	Option 4
(c)	3.5.1	Drawings, engineering and associated lists level 3, (production)	DI-E-7015	Design activity designation - by contractor Drawing no. - by contractor Delivery of hard copy - to procuring activity
(d)	3.5.3	Manual, technical preliminary	DI-M-2043	MIL-M-15071, type I
(e)	3,5.3	Manuals, camera ready	DI-M-2044	MIL-M-15071, type I
	4.1.1	Inspection system	DI-R-4B03	-------

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(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed , by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.3 The data requirements of 6.2.2 and any task in section 3, 4, or 5 or the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes,

Preparing activity:
Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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